KAZAKH BRITISH TECHNICAL UNIVERSITY

Data collection and preparation

SIS

Climate and City Size in Kazakhstan

API and web scraping, cleaning, merging, EDA and visualization

Checked by	Dina Koishiyeva
Done by	Tursunbekov Mukhit
Group time	Thursday, 12:00–13:00
ID	23B031160

Almaty, 2026

Objective

To apply the complete data collection and preparation pipeline: collect data from an API and from a static web page, clean, merge and transform it, perform basic exploratory analysis and communicate the insights with visualizations. **Research question:** is the climate of Kazakhstan's largest cities (temperature, precipitation, elevation) related to their population?

1. Data collection

API. Open-Meteo Geocoding API (coordinates and elevation of each city) and Open-Meteo Historical Weather API (daily mean temperature and precipitation for 2023), called with *requests*.

Web scraping. The Wikipedia page *List of most populous cities in Kazakhstan* was downloaded after a robots.txt check and parsed with *BeautifulSoup*: 87 rows, column *Population (January 2023 estimate)*.

2. Data cleaning

Scraped table: footnotes removed, population converted from text to integers, columns renamed to snake_case; no duplicates or missing values; the 15 most populous cities were kept. **Weather:** 5,475 daily rows (15 cities × 365 days), dates converted to datetime, no missing values or duplicates (the code still handles gaps by interpolation for temperature and 0 mm for precipitation).

3. Merging

Daily weather was aggregated to one row per city (mean, min, max and standard deviation of temperature, total precipitation, days with at least 1 mm) and joined with population and geography on the city name. All 15 cities matched; the result has 12 columns.

4. Analysis

Variable	Mean	Minimum	Maximum
Population	587,089	222,636 (Petropavl)	2,161,797 (Almaty)
Mean annual temperature, °C	9.1	5.0 (Petropavl)	14.7 (Shymkent)
Annual precipitation, mm	461	110.6 (Kyzylorda)	760.0 (Oskemen)
Elevation, m	273	−19 (Atyrau)	783 (Almaty)

Correlation with population (Pearson, n = 15): elevation **0.74**, latitude −0.40, mean temperature 0.18, precipitation 0.18, temperature range −0.17, rainy days 0.07. Latitude and mean temperature: −0.91.

5. Visualization

Five plots were made with Matplotlib and Seaborn (next page): mean temperature by city, population vs temperature, correlation matrix, monthly temperature of the five largest cities, latitude vs temperature.

Conclusions

1. **Continental climate.** Every city has a gap of 47 to 68 °C between its coldest and warmest day, and temperature falls steadily with latitude. Shymkent is the warmest city, Petropavl the coldest, Oskemen the wettest, Kyzylorda the driest.
2. **Temperature and precipitation do not explain city size** (r = 0.18 for both): large cities exist in warm and in cold climates.
3. **Elevation stands out** (r = 0.74); without Almaty it is still 0.55, and the rank correlation is 0.76. It is most likely a stand-in for region (southern and central cities lie on higher ground), not a direct cause.
4. **Limitations.** Only 15 cities, one year of weather and one population snapshot; city size is driven mostly by economics and politics. Next step: more years and cities, plus economic variables.

Visualizations

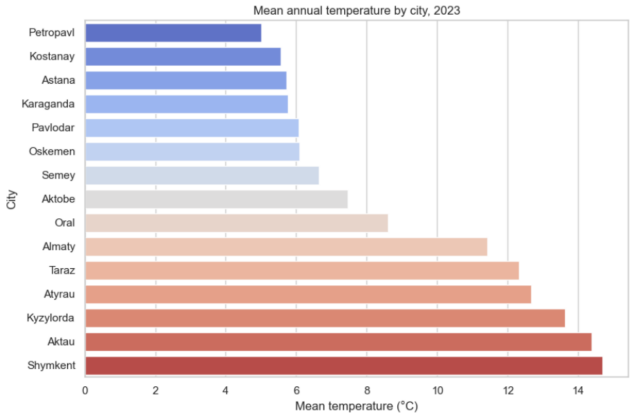


Figure 1. Mean annual temperature by city, 2023.

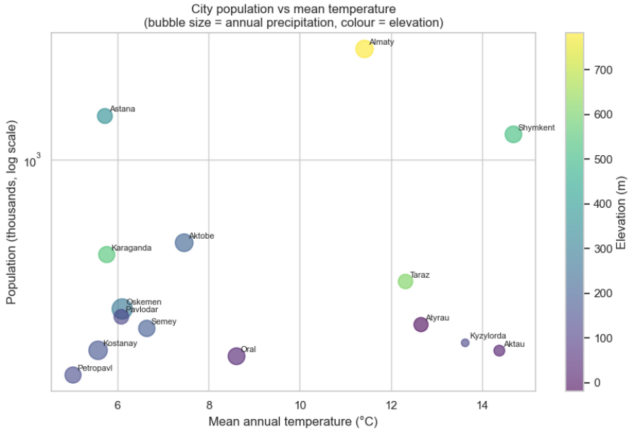


Figure 2. Population vs mean temperature (size = precipitation, colour = elevation).

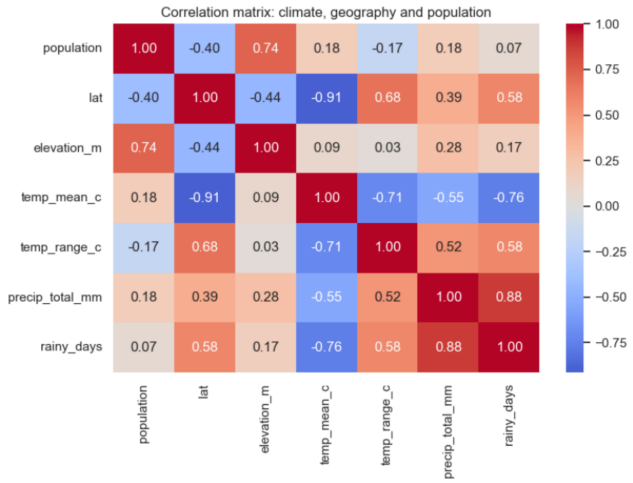


Figure 3. Correlation matrix: climate, geography and population.

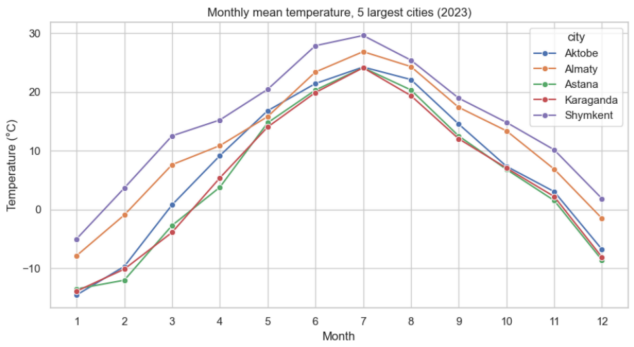


Figure 4. Monthly mean temperature, 5 largest cities, 2023.

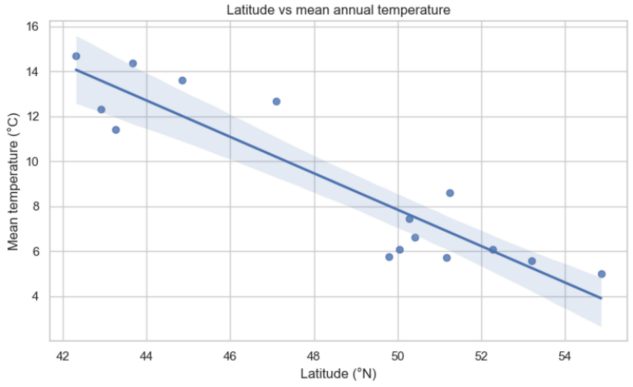


Figure 5. Latitude vs mean annual temperature.

Reading the plots

Fig. 1 and 4: the south is warmer, but all cities have cold winters and hot summers.

Fig. 3: population is tied mostly to elevation (0.74), not to temperature (0.18).

Fig. 5: latitude explains most of the temperature differences (r = −0.91).